Mathematics · Practice Test 1

40 questions · Calculator allowed · Use the Grade 8 Mathematics Reference Sheet

Read each question carefully. Watch for directions like *Select TWO* and *least to greatest*.

Reporting Category 1 · Numerical Representations and Relationships

1. Which list is ordered from **least to greatest**?

A. ⅔, 68%, 0.7, √0.5 B. √0.5, 0.7, 68%, ⅔ C. 68%, ⅔, √0.5, 0.7 D. 0.7, √0.5, ⅔, 68%

2. Between which two whole numbers does √40 lie?

A. 5 and 6 B. 6 and 7 C. 19 and 21 D. 20 and 21

3. *(Griddable)* Write 68,000 in scientific notation. ☐

4. Which number is **irrational**? A. √36 B. ⅘ C. π D. 0.25

Reporting Category 2 · Computations and Algebraic Relationships

5. What is the slope of the line through (1, 2) and (4, 11)? A. 2 B. 3 C. 6 D. 9

6. A proportional relationship passes through the origin and the point (3, 12). What is the unit rate? A. 3 B. 4 C. 9 D. 15

7. Why is the slope of a straight line the same between any two points?

A. The line is horizontal B. The right triangles formed are **similar**, so rise÷run is constant C. All lines have slope 1 D. The points are always 1 unit apart

8. *(Equation editor)* A plant is 6 cm tall and grows 3 cm per week. Write an equation for its height y after x weeks. ☐

9. Which set of ordered pairs represents a **function**?

A. {(1, 2), (1, 3), (2, 4)} B. {(2, 5), (3, 5), (4, 5)} C. {(0, 1), (0, 2), (0, 3)} D. {(5, 1), (5, 2), (6, 3)}

10. Which equation represents a **proportional** relationship? A. $y = 4x$ B. $y = 4x + 2$ C. $y = x - 4$ D. $y = 2x - 1$

11. A gym charges a $15 membership fee plus $6 per visit. Which equation gives the cost y for x visits? A. $y = 15x + 6$ B. $y = 6x + 15$ C. $y = 21x$ D. $y = 6x - 15$

12. *(Griddable)* For a direct variation, $y = 20$ when $x = 5$. What is y when $x = 9$? ☐

13. Which table shows a **non-proportional** relationship?

A. (1, 3)(2, 6)(3, 9) B. (1, 5)(2, 10)(3, 15) C. (1, 4)(2, 7)(3, 10) D. (2, 8)(3, 12)(4, 16)

14. Which situation is **proportional**?

A. A taxi charges $3 plus $2 per mile B. A pool has 10 L and drains 2 L per minute C. Apples cost $2 per pound D. A phone plan is $20 plus $0.10 per text

15. *(Griddable)* Solve for x: $6x + 4 = 2x + 20$. ☐

16. "Three less than twice a number equals the number plus five." Which equation models this? A. $2x - 3 = x + 5$ B. $3 - 2x = x + 5$ C. $2x + 3 = x - 5$ D. $3x - 2 = x + 5$

17. Which situation can be modeled by $5x + 10 = 3x + 30$?

A. Two savers start with $10 and $30 and save $5 and $3 per week; when are the totals equal? B. A number times 5 C. The area of a rectangle D. The sum of 10 and 30

18. Two lines are graphed and cross at (2, 5). What is the solution of the system? A. $x = 5, y = 2$ B. $x = 2, y = 5$ C. $x = 7$ D. No solution

19. *(Select TWO)* Which statements about the line $y = 2x - 3$ are true? A. The slope is 2 B. The slope is −3 C. The y-intercept is −3 D. It passes through the origin

Reporting Category 3 · Geometry and Measurement

20. A cylinder has radius 4 and height 7. What is its volume? ($\pi \approx 3.14$) A. 87.92 B. 175.84 C. 351.68 D. 703.36

21. A cone has radius 6 and height 10. What is its volume? ($\pi \approx 3.14$) A. 125.6 B. 376.8 C. 1130.4 D. 3768

22. A cylinder has radius 3 and height 8. What is its **total** surface area? ($\pi \approx 3.14$) A. 150.72 B. 169.56 C. 207.24 D. 263.76

23. *(Griddable)* A right triangle has legs of 7 and 24. What is the length of the hypotenuse? ☐

24. What is the distance between (−1, −3) and (3, 0)? A. 5 B. 7 C. 12 D. 25

25. The point (−2, 5) is dilated by a scale factor of 3 about the origin. What is the image? A. (−6, 15) B. (6, 15) C. (−5, 8) D. (1, 8)

26. The point (5, −4) is reflected over the y-axis. What is the image? A. (5, 4) B. (−5, −4) C. (−5, 4) D. (4, −5)

27. The point (2, 7) is rotated 90° counterclockwise about the origin. What is the image? A. (7, 2) B. (−7, 2) C. (7, −2) D. (−2, −7)

28. Which transformation does **not** preserve congruence? A. Translation B. Reflection C. Rotation D. Dilation (scale factor 3)

29. The angles of a triangle measure 3x, 2x, and x. What is the measure of the largest angle? A. 30° B. 60° C. 90° D. 120°

30. Two similar rectangles have corresponding sides in a ratio of 2 : 5. The smaller has a length of 6. What is the matching length of the larger? A. 10 B. 12 C. 15 D. 18

31. In the cylinder volume formula $V = Bh$, what does B represent? A. $2\pi r$ B. πr^2 C. πd D. $r + h$

32. A square with side 4 is dilated by a scale factor of 3. What is the area of the image? A. 12 B. 48 C. 108 D. 144

Reporting Category 4 · Data Analysis and Personal Financial Literacy

33. *(Griddable)* Find the mean absolute deviation of 3, 5, 7, 9. ☐

34. A trend line is modeled by y = 1.5x + 2. Predict *y* when x = 8. A. 11 B. 12 C. 14 D. 16

35. On a scatterplot, the points fall from the upper-left to the lower-right. What association is shown? A. Positive linear B. Negative linear C. No association D. Non-linear

36. Which description suggests a **linear** association between two variables? A. Points scattered randomly B. Points following a straight upward path C. Points forming a U-shape D. Points in a circle

37. *(Griddable)* Find the simple interest on $800 at 5% for 4 years. ☐

38. For the same loan amount and interest rate, which loan costs the **most** total interest? A. A 2-year loan B. A 3-year loan C. A 5-year loan D. They cost the same

39. How do small amounts of money invested regularly behave over time? A. They lose value B. They stay exactly the same C. They grow, especially with compound interest D. They cannot be saved

40. College costs $12,000. A family expects to contribute $4,000 and wants to save the rest over 16 months. How much must they save each month? A. $250 B. $500 C. $750 D. $1,000

Answers and full explanations for this test are in Part 3 · Answer Keys & Explanations.